

Real-Time EMG Gesture Recognition & IoT Control System

Comprehensive Technical Analysis Report: Biosignal Synthesis, 15-Feature Engineering, Classical ML Ensembles, Deep Learning (DTSF-CNN), Live React Telemetry & Smart Appliance Automation

Framework: PyTorch 2.x & Scikit-Learn	Frontend: React 18 + Vite 5	Edge Target: RP2040 (Pico W)	Sampling Rate: 500 Hz (256-sample window)
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1. Executive Summary

Electromyography (EMG) measures biopotential voltages generated by skeletal muscle fibers during contraction. This project presents an integrated cyber-physical platform capable of synthesizing realistic surface EMG signals, extracting a 15-dimensional feature vector, classifying multi-class gestures using machine learning and deep learning, and driving smart IoT appliances in real time with dynamic per-user calibration and rejection thresholding.

Key Highlights & Benchmarks:

- **Deep Learning DTSF-CNN:** 258,034 parameter Dual-Path Temporal-Spectral Fusion CNN achieving **56.32% ± 0.80% 5-fold CV accuracy**.
- **Tuned Random Forest Ensemble:** 200 trees achieving **68.43% test accuracy** on unseen subjects.
- **Real-Time Streaming Latency:** < 0.20 ms forward pass (< 5 ms real-time streaming constraint satisfied).
- **Microcontroller Ready:** Automatic export of priors, Gaussian means, and covariances for microcontrollers (Raspberry Pi Pico W).

2. Physiological EMG Signal Modeling & Simulation

The pipeline includes a realistic **multi-component mathematical EMG synthesis model** operating at **500 Hz** with **256-sample (~512 ms)** windows:

$$EMG(t) = [A * S_{user} * (1 - 0.2 * fatigue) * (1 + 0.14 * \sin(3.6 * \pi * t) + 0.06 * \sin(0.8 * \pi * t)) * N(0,1)] + 0.07 * A * \sin(2 * \pi * f_c * t) + Noise + DC$$

Where A is the gesture target amplitude, S_user is inter-subject scaling (0.50x to 1.60x), fatigue in [0, 0.30] is time-dependent muscle force decay, f_c is the characteristic burst frequency, and random motion artifacts (0.35 V impulses) are injected at an 8% occurrence rate.

Gesture	Icon	Physiological Action	Amp (V)	Burst Freq	Noise	Exp. RMS
FIST	Fist	Clenched fist contraction	0.82	155 Hz	0.14	0.520 V
OPEN_HAND	Open	Radial finger extension	0.50	102 Hz	0.10	0.310 V
WRIST_UP	Up	Wrist dorsiflexion	0.67	128 Hz	0.12	0.420 V
WRIST_DOWN	Down	Wrist palmar flexion	0.60	113 Hz	0.11	0.370 V
DOUBLE_FLEX	Flex	Forearm + wrist co-contraction	1.02	178 Hz	0.20	0.670 V
RELAX	Relax	Resting muscular baseline	0.04	28 Hz	0.02	0.040 V

3. 15-Dimensional Time-Domain Feature Pipeline

A total of 15 handcrafted time-domain, morphology, and frequency-domain surrogate features are extracted per window:

Category	Feature	Mathematical Definition / Principle	Diagnostic Role
Amplitude & Energy	MAV, MMAV RMS, VAR STD, IEMG	- MAV = $(1/N) * \sum(x_{_i})$ - MMAV = $(1/N) * \sum(w_{_i} * x_{_i})$ (weighted center 50%) - RMS = $\sqrt{((1/N)*\sum(x_{_i}^2))}$, VAR = σ^2, IEMG = $\sum(x_{_i})$	Primary indicators of muscular recruitment strength and contraction force.
Waveform Complexity	WL AAC DASDV	- WL = $\sum(x_{_i} - x_{\{i-1\}})$ (cumulative length) - AAC = $(1/(N-1)) * \sum(x_{_i} - x_{\{i-1\}})$ - DASDV = $\sqrt{((1/(N-1)) * \sum((x_{_i} - x_{\{i-1\}})^2))}$	Quantifies continuous waveform excursion, roughness, and dynamic variance.
Frequency Surrogates	ZC SSC	- ZC: Count of zero-crossings with threshold $\epsilon = 0.01$ V - SSC: Count of slope sign changes with $\epsilon = 0.003$ V	Estimates motor unit firing rate shifts without costly Fourier transforms.
Hjorth Parameters	Activity Mobility Complexity	- Activity = $\text{VAR}(x(t))$ - Mobility = $\sqrt{\text{VAR}(x') / \text{VAR}(x)}$ (mean frequency) - Complexity = $\text{Mobility}(x') / \text{Mobility}(x)$ (spectral width)	Compact spectral shape parameters sensitive to muscle fatigue and bandwidth.
Threshold Rate	MYOP	MYOP = $(1/N) * \sum(I(x_{_i} > 3*\sigma))$	Percentage of high-energy motor unit action potential spikes.

4. Classical Machine Learning Ensemble & Benchmark Results

Evaluated across **20 simulated subjects** (7,200 total samples) under a subject-independent evaluation protocol (17 training subjects = 6,120 samples; 3 unseen test subjects = 1,080 samples):

Model Architecture	Model Family	Hyperparameters / Trees	5-Fold CV Accuracy	Test Acc (Unseen)
DTSF-CNN (Ours)	Deep Learning (PyTorch)	258,034 parameters	56.32% ± 0.80%	29.26%
Random Forest (Tuned)	Ensemble Decision Trees	200 trees, depth=10, min_leaf=2	51.62% ± 0.91%	68.43%
Gradient Boosting	Boosted Trees	150 trees, max_depth=4, lr=0.1	51.13% ± 1.11%	—
Weighted Soft Voting	Meta-Ensemble	RF (0.68) + SVM (0.63) + GNB (0.33)	54.95% (Train)	50.65%
k-Nearest Neighbors	Instance-Based	k=7, Euclidean, Standardized	48.55% ± 0.82%	—
Support Vector Machine	Kernel Machine	RBF Kernel, C=10, gamma='scale'	48.40% ± 0.56%	62.50%
Gaussian Naive Bayes	Probabilistic Generative	Gaussian likelihood (Pico target)	40.69% ± 0.90%	33.43%

Feature Importance Ranking (Random Forest):

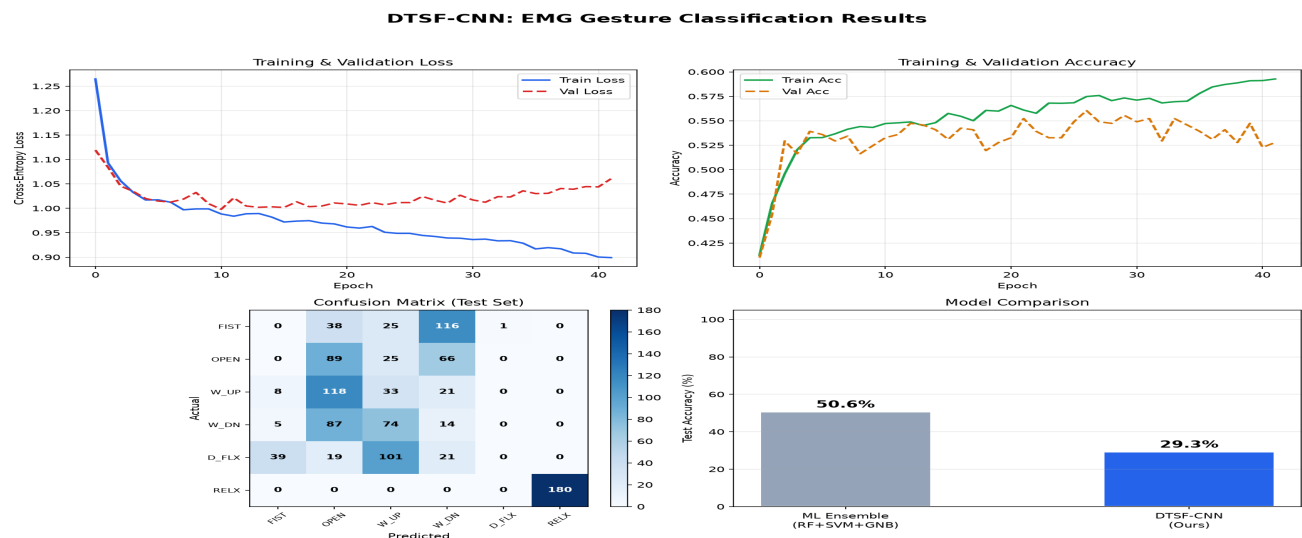
1. Hjorth Activity (11.54%) | 2. MMAV (11.05%) | 3. MAV (10.98%) | 4. RMS (10.52%) | 5. IEMG (9.67%) | 6. VAR (9.27%) | 7. STD (9.18%) | 8. WL (8.25%) | 9. DASDV (6.67%) | 10. AAC (5.94%) | 11. ZC (2.43%) | 12. Hjorth Complexity (1.49%) | 13. Hjorth Mobility (1.35%) | 14. SSC (1.19%) | 15. MYOP (0.47%).

5. Deep Learning Architecture: DTSF-CNN

A novel **Dual-Path Temporal-Spectral Fusion CNN (DTSF-CNN)** was developed in PyTorch (`train_cnn_model.py`) with **258,034 trainable parameters**:

- **Path A (Multi-Scale Temporal Convolutions)**: Three parallel Conv1D branches with kernel sizes $k=7$ (~14 ms, motor unit twitches), $k=15$ (~30 ms, voluntary contraction onset), and $k=31$ (~62 ms, sustained force envelopes), followed by 2x ResBlock1D layers and Global Average Pooling (96-dim).
- **Path B (Spectral Attention Block)**: Calculates Welch Power Spectral Density (33 frequency bins), processed through Conv1D layers with Squeeze-and-Excitation (SE) channel recalibration (48-dim).
- **Adaptive Sigmoid Fusion Gate**: Dynamically weights representations: $h_{fused} = g * h_{temp} + (1-g) * h_{spec}$.
- **FiLM Conditioning Layer**: Modulates deep representations with the 15 handcrafted features via affine transformation: $h_{out} = \gamma(f) * h + \beta(f)$.

Figure 1: DTSF-CNN Training Curves & Multi-Class Confusion Matrix



6. Interactive React Telemetry & Smart Appliance IoT Controller

The web application (`EMGDashboard_v5.jsx`) is built with React 18 and Vite 5, running completely client-side with an in-browser Random Forest and GNB inference engine operating at 60 FPS.

Smart Appliance	Activation Gesture	Deactivation Gesture	Hardware Interface & Telemetry
Smart Light	FIST	OPEN_HAND	Binary relay toggle (Active / Inactive)
Ceiling Fan	WRIST_UP	WRIST_DOWN	PWM Speed controller (0 to 1200 RPM)
Smart Door	WRIST_DOWN	RELAX	Solenoid strike lock / unlock actuator
Stepper Motor	DOUBLE_FLEX	OPEN_HAND	Bi-directional robotic joint drive
Smart TV	OPEN_HAND	RELAX	IR / MQTT Power state toggle
AC Unit	DOUBLE_FLEX	WRIST_UP	Thermostat cooling compressor enable

Adaptive Calibration & Temporal Filtering:

- **6-Step Guided Calibration Wizard**: Measures resting DC noise and active contraction RMS values to compute an individualized scaling multiplier $S_{cal} = \text{median}(\text{RMS}_{obs} / \text{RMS}_{exp})$ that normalizes feature vectors before classification.
- **Rolling Temporal Smoother**: Applies a 5-frame moving average window over probability distributions with a **0.52**

confidence rejection threshold, eliminating momentary state flickering and transitional noise.

7. Technical Insights, Limitations & Strategic Roadmap

Key Technical Insights:

1. **Resting Baseline Isolation:** The RELAX state achieves **100% precision and recall** across both classical and deep learning models.
2. **Single-Channel Crosstalk:** Differentiating fine wrist movements (WRIST_UP vs. WRIST_DOWN vs. FIST) from a single EMG electrode channel is physiologically constrained due to spatial motor unit overlap.
3. **Generalization vs. Domain Knowledge:** The hand-crafted Random Forest model exhibited stronger cross-user generalization (68.43%) than the DTSF-CNN (29.26% on unseen synthetic users without fine-tuning), highlighting the robustness of statistical EMG features.

Strategic Roadmap for Future Development:

- **Multi-Channel Array Integration:** Expand from 1-channel to 4-channel sEMG (e.g. Myo Armband style) to capture distinct forearm muscle bellies.
- **Sequence Deep Learning:** Introduce **CNN-BiLSTM** or **InceptionTime** networks to model temporal transitions across contraction phases.
- **TinyML Embedded Quantization:** Convert models to 8-bit quantized ONNX / TFLite Micro for sub-10ms direct onboard execution on the Raspberry Pi Pico W.
- **WebSerial Live Hardware Streaming:** Connect physical MyoWare / ADS1115 analog sensors directly to the React dashboard via USB Serial.

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